ANSWER KEY & MARKING SCHEME SET 1

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** cell	All living things are made of cells.
2	**(d)** tissue	Tissue.
3	**(a)** cell, tissue, organ	Small to large.
4	**(c)** cork	Hooke observed cork.
5	**(d)** organ	Organ.
6	**(d)** rectangular	Due to the cell wall.
7	**(a)** nucleus	The nucleus controls the cell.
8	**(a)** cytoplasm	It fills the cell.
9	**(d)** no chloroplasts	Root and bulb cells are underground.
10	**(a)** chloroplast	Gives green colour.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Smallest structural and functional unit of life (2).	2
12	Contains chlorophyll (1); Helps in making food by photosynthesis (1).	1+1
13	Yeast (1); Mushroom (1).	1+1
14	Cell wall (1); Chloroplast (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Cell wall is present in plant cell only (1); Chloroplasts are in plant cells only (1); Plant cell has a large vacuole (1).	1+1+1
16	Human (1); Mango tree (1); Earthworm (1).	1+1+1
17	Functions differ (1); Shape suits function (1); e.g. nerve cell long, RBC round (1).	1+1+1
	OR: Nucleus: control (1); Cell membrane: protects and regulates movement (1); Cytoplasm: cell activities (1).	1+1+1
18	To see cells clearly (1); Stain gives contrast (1); Safranin or iodine (1).	1+1+1
	OR: Soil (1); Water (1); Inside our body (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cell membrane: outer covering (1); Cytoplasm: jelly-like (1); Nucleus: control (1); No cell wall (1); No chloroplast (1).	1+1+1+1+1
	OR: Scrape the cheek gently (1); Place on a slide with water (1); Stain (1); Cover with a cover slip (1); Observe and draw (1).	1+1+1+1+1
20	Cell wall (1); Chloroplast (1); Vacuole (1); Shape (1); Nutrition (1).	1+1+1+1+1
	OR: Unicellular: one cell (1); Multicellular: many cells (1); Examples: amoeba, yeast (1); Examples: tree, human (1); In multicellular organisms, cells divide the work (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.
- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 2

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** all cells	Every cell has one.
2	**(c)** an animal cell	From the human mouth.
3	**(d)** plant cell	Only plants have walls.
4	**(b)** fungus	It is a single-celled fungus.
5	**(a)** see parts clearly	Contrast.
6	**(d)** unicellular	Single cell.
7	**(a)** fungus	Lacks chlorophyll.
8	**(c)** Hooke	In 1665.
9	**(b)** human	Many cells.
10	**(d)** bacteria	Lactobacillus.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Controls the activities of the cell (1); Holds genetic material (1).	1+1
12	Useful: Lactobacillus forms curd (1); Harmful: causes cholera (1).	1+1
13	Organism made of a single cell (1); Amoeba (1).	1+1
14	Microscope (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Bacteria (1); Fungi (1); Protozoa (also viruses) (1).	1+1+1
16	Take a thin peel (1); Place it on a slide with stain and cover slip (1); Observe under the microscope (1).	1+1+1
17	Magnification: enlargement of image (1); 10 x 10 (1); 100x (1).	1+1+1
	OR: Cell to tissue (1); Tissue to organ (1); Organ to organ system to organism (1).	1+1+1
18	Tissue is made of similar cells (1); Organ is made of different tissues (1); Organ performs bigger functions (1).	1+1+1
	OR: Onion peel: plant cells (1); Cheek cells: animal cells (1); Bacteria: from curd or water (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cell: nerve cell (1); Tissue: muscle (1); Organ: stomach (1); Organ system: digestive system (1); Organism: human (1).	1+1+1+1+1
	OR: Tiny non-cellular particles (1); Multiply only in living host cells (1); Cause colds, influenza, COVID (1); Not killed by antibiotics (1); Vaccines protect (1).	1+1+1+1+1
20	Observed cork under a microscope (1); Saw box-like compartments (1); Called them cells (1); Year was 1665 (1); He was not able to see the parts inside (1).	1+1+1+1+1
	OR: Vacuole: storage (1); Chloroplast: food making (1); Cell wall: support (1); Nucleus: control (1); Cytoplasm: site of activities (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.
- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 2: **Invisible Living World: Beyond Our Naked Eye** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** bacterium	Microscopic.
2	**(d)** microscope	It enlarges small things.
3	**(b)** protozoan	Plasmodium.
4	**(c)** virus	Acellular.
5	**(d)** epidermis	It is used for slides.
6	**(a)** virus	Viruses are parasites.
7	**(d)** protozoan	Single-celled.
8	**(c)** single-celled	Microscopic organisms.
9	**(c)** vacuole	Large central vacuole.
10	**(b)** near the eye	You look through it.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Group of tissues performing a specific function (1); Heart, lungs (1).	1+1
12	Organism made of many cells (1); Human (1).	1+1
13	Group of similar cells doing a common job (1); Muscle tissue (1).	1+1
14	They do not have cell structure (1); They have genetic material in a protein coat (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	To see cells (1); To see bacteria and protozoa (1); To study diseases (1).	1+1+1
16	Yeast: bread (1); Lactobacillus: curd (1); Penicillium: penicillin (1).	1+1+1
17	Eyepiece (1); Objective lens (1); Stage and mirror (1).	1+1+1
	OR: Amoeba (1); Paramecium (1); Yeast (1).	1+1+1
18	Diseases (1); Spoiling of food (1); Damage to crops (1).	1+1+1
	OR: Hold with both hands (1); Do not touch lenses (1); Keep it in a box after use (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Bacteria: single-celled (1); Bacteria: some cause diseases (1); Fungi: no chlorophyll (1); Fungi: yeast and mushrooms (1); Both are decomposers (1).	1+1+1+1+1
	OR: Nerve cell (1); RBC (1); Muscle cell (1); Root hair cell (1); Shape and function fit (1).	1+1+1+1+1
20	Yeast: bread (1); Lactobacillus: curd (1); Penicillium: antibiotic (1); Vibrio cholerae: cholera (1); Plasmodium: malaria (1).	1+1+1+1+1
	OR: Eyepiece and objective lens (1); Magnification = product (1); E.g. 10x and 40x give 400x (1); Used to see cells (1); Handle with care (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that animal cells also have a cell wall or chloroplasts.
- Confusing the **cell wall** (outer, rigid, plants) with the **cell membrane** (inside the wall, in all cells).
- Believing cork contains living cells, when Hooke saw the walls of dead cells.
- Calculating total magnification by adding the lens powers instead of multiplying.
- Writing that all cells have a nucleus; bacteria have no true nucleus.
- Forgetting the stain, thin peel and cover slip when describing slide preparation.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.